Sphere

Sphere Forge: from description and drawings to CAD

Build an AI agent that turns a design brief and technical views into a precise CAD model.

The challenge

In

Design brief

Written description with exact dimensions

→

In

8 views



Plus the part in its assembly

→

Agent

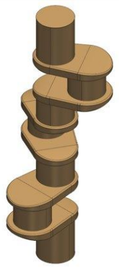
Read, plan, code

Writes parametric CAD , checks against the views, refines

→

Out

CAD solid



One STEP file per part, in mm

→

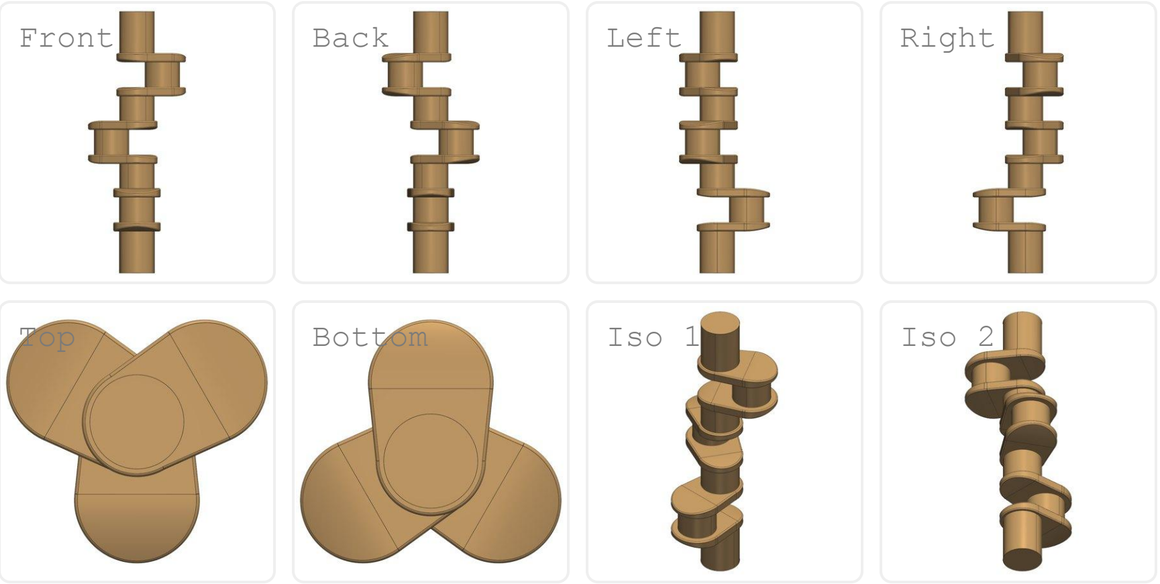
Score

0 → 1

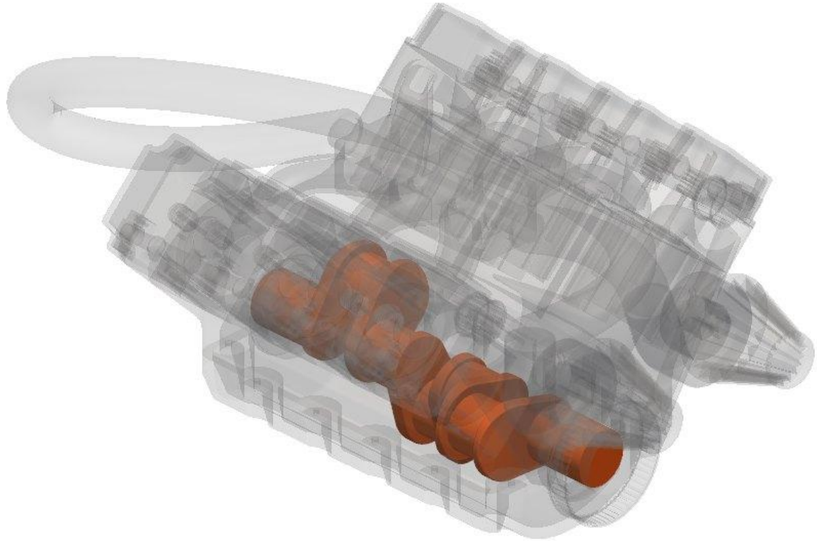
Scored automatically

Position and orientation don't matter.
Shape, size, holes and handedness do.

What an agent receives



8 orthographic views, all in the part's own X/Y/Z frame, so text and images agree

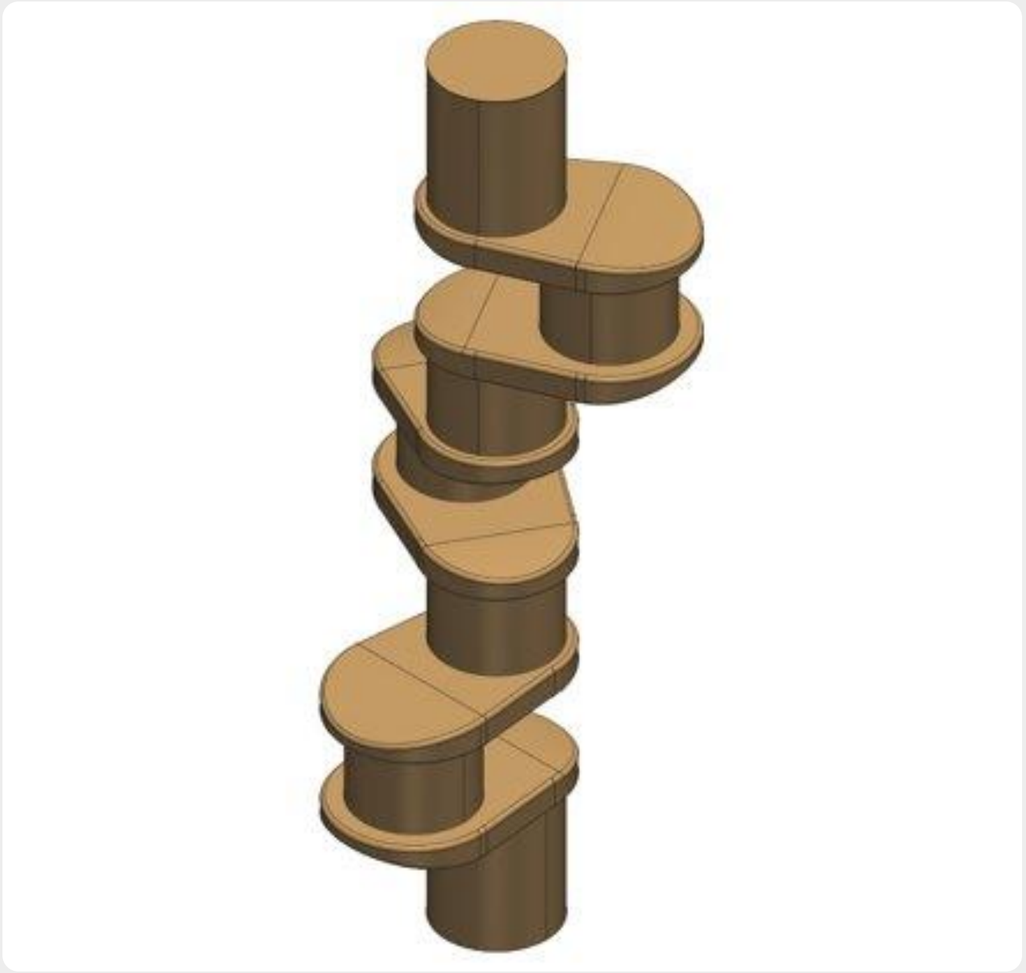


The part highlighted in its engine assembly, to show its function

Design brief · every number measured from the CAD model

Three-throw crankshaft. Envelope 211.59 × 196.85 × 564.88 mm; volume 3472 cm³. 3 crank pins, Ø101.6 mm, each offset 63.5 mm from the rotation axis, 120° apart.

What teams hand in

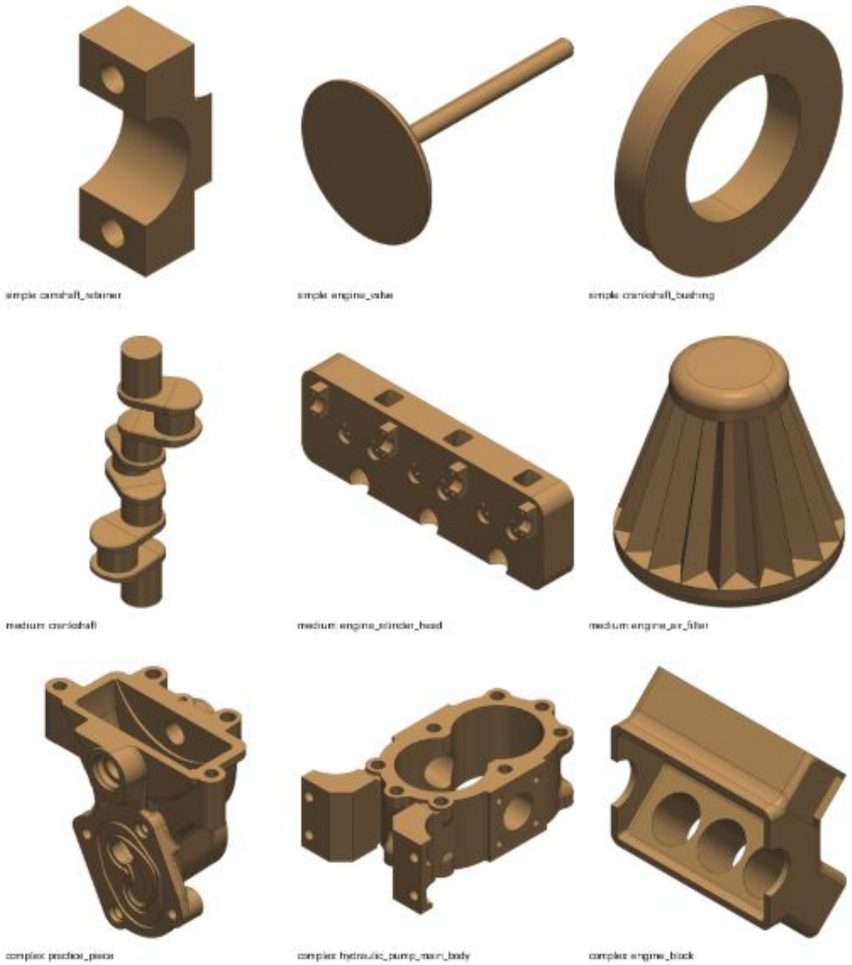


One solid per part

`outputs/<id>.step`

0 A missing, broken or non-solid file scores 0 for that part. The rest still counts.

The parts



9

Practice parts, released with the answers

9

Hidden parts, never released

3

Difficulty tiers, weighted equally

8

Views per part

`Practice: 3 simple · 3 medium · 3 complex`

`Tiers: ≤ 30 faces simple · ≤ 200 medium · more is complex`

How we score (1h of run time for the nine parts allowed)

`score = 0.30 · volume + 0.30 · surface + 0.20 · dimensions + 0.20 · holes`

30 %

Volume overlap

Is the overall form right?

30 %

Surface match

Is the surface in the right place, and how precisely?

`Checked at 0.5–10 % of part size`

20 %

Dimensions

Is it the right size?

`2 % off loses 10 %; 20 % off loses all`

20 %

Holes

Are all the through-holes there?

`Counted exactly from CAD topology`

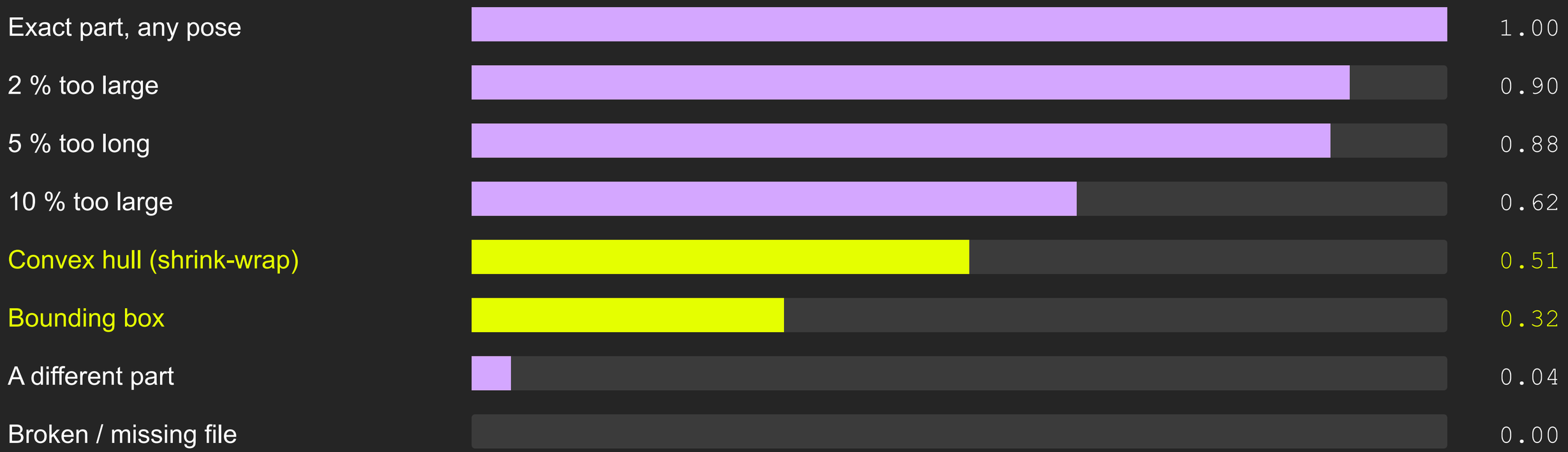
`Alignment first`

Rotation and translation only. No scaling, no mirroring.

`Same scorer for teams`

Shipped with the practice set, so local scores match the leaderboard.

What scores mean



Shortcuts don't pay.

A bounding box scores about 0.3, while a part only 2 % too large still scores 0.9.

Seeing the score

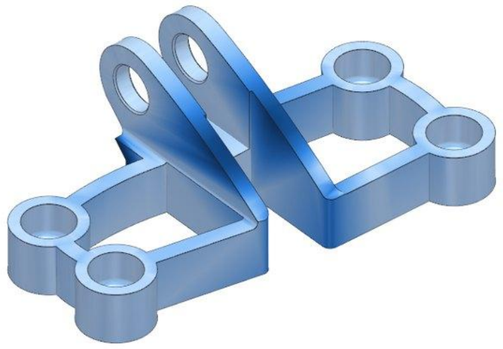
0.91 **Close match**

0.7 mm average deviation, every hole present

Original



Design, by deviation



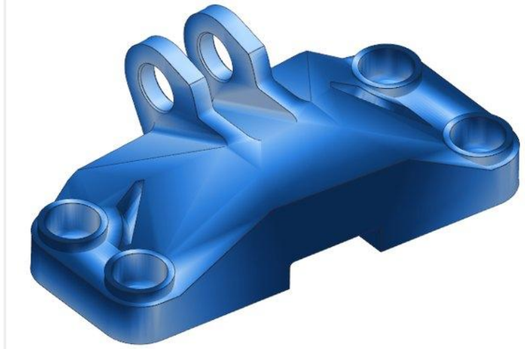
0.56 **Poor match**

77 % too much material, 6 of 10 holes

Original



Design, by deviation



0 mm — 5 mm

Distance of each point to the original surface, after alignment

Event timeline

During the hackathon

Live scores

Upload STEP files; they are scored live on the leaderboard

→

Friday 08:00

Code freeze

Teams upload their pipeline

→

Evaluation

Hidden scoring

Pipelines run on the hidden parts with our script

→

Final

Top 4 present

10 minutes each, including questions. The jury picks the winner.

While pipelines are evaluated

Every team prepares 3 simple slides about its solution.

Live scoring during the hackathon

01

Team repo

From the submission template

→

02

Upload parts

STEP files only

→

03

Automatic scoring

Picked up within ~30 s; seconds to a minute per part

→

04

Leaderboard

Updated live

Ranking

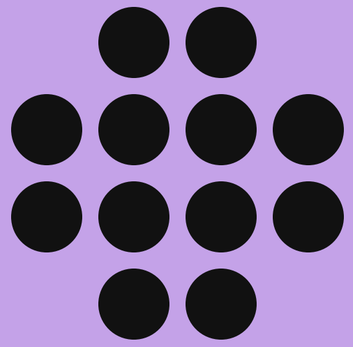
Each team's **best** commit counts, not its latest.

Isolation

Scoring is isolated per part: a file that crashes the CAD kernel can't affect anyone else.

Code freeze: Friday 08:00

Teams upload their pipeline. Each pipeline is run on the hidden parts and evaluated with our scoring script.



Final presentations

3

Simple slides per team

4

Best teams on the hidden score present to the jury, 10 minutes each with questions

Your slides answer

01	What solution did you build?
02	What makes it special?
03	Is it scalable?
04	How could it be improved?